

A Diagnostic Evaluation of Small Language Models for Offline Writing-Feedback Tutoring in Brazilian Portuguese: A Study on Structural and Pedagogical Adherence under Public-School Infrastructure Constraints

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Abstract

The adoption of generative Artificial Intelligence in education has so far relied largely on proprietary models that depend on broadband connectivity, an assumption incompatible with the material reality of most Brazilian public schools. This work conducts a controlled diagnostic evaluation of eight open Small Language Models (SLMs), executed strictly in a local, offline environment, against a metalinguistic rubric derived from Ingedore Koch's Textual Linguistics. We performed 312 primary inference calls (8 models $\times$ 13 canonical scenarios $\times$ 3 repetitions) without any external grammatical coercion mechanism, together with 80 additional calls across four falsification isolations directed at the study's most salient architectural finding. Categorical comparisons rely on Fisher's exact test (with odds ratio and 95% CI). The results reveal two distinct failure regimes. The first is a zero-shot structural non-adherence, observed as 100% typological divergence in the Llama 3.2 family (1 B and 3 B) and in Phi-3 Mini. For the Llama 3.2 family this divergence is a deterministic formatting bias inherited from instruction tuning for generalist commercial use, and it proves fully reversible by a minimal in-context demonstration (*one-shot*), which sets it apart from genuine compositional incapacity. The second regime, treated here as a corroborating qualitative probe rather than a co-equal claim, is pedagogical superficiality in Qwen 2.5 3B-Instruct, the only structurally conformant model in the deployable range under zero-shot: across all three repetitions its responses mobilize Koch's rubric terminology in only about half of the cases (51.3%), and they do so unstably, with a single scenario out of thirteen eliciting that mobilization in every repetition. To ground the pedagogical construct in expert practice, we complement the benchmark with a specialist human reference: 65 feedback turns produced by five experienced teachers over the same scenarios, coded into eight mediation functions that emerge from the corpus (inter-annotator $\kappa = 0.83$). This reference is used descriptively, not as a performance yardstick against the model; it characterizes the repertoire of expert mediation, anchors the rubric, and defines the pedagogical target a specialized model would have to reproduce. Coding the conformant model's own outputs against these eight functions makes the gap concrete: its mediation concentrates on format-forced recognition and open expansion (FM01, FM07) while almost entirely missing the expert corrective core of naming the problem, offering a cue, and proposing a revision (FM02/FM04/FM06: 3%/15%/0% against 80%/69%/57%). We conclude that contemporary off-the-shelf generalist models are insufficient for fine-grained writing-feedback tutoring in Brazilian Portuguese under a zero-shot regime, a result that motivates the development of specialized models through supervised fine-tuning on curated pedagogical corpora.

Keywords: Language Models; Basic Education; Brazilian Portuguese; Local Inference; Textual Linguistics; Diagnostic Evaluation

1 Introduction

Over the past three years, the incorporation of Large Language Models (LLMs) into education has become one of the fastest-growing research fronts in Computers and Education. The specialized literature documents promising results with proprietary models such as GPT-4, Claude, and Gemini for conversational tutoring, automated essay scoring, and formative *feedback* generation (Kasneci et al., 2023; Tlili et al., 2023). This body of work, however, rests largely on infrastructural assumptions, such as stable broadband access, the absence of budget limits on Application Programming Interface (API) calls, and the contractual stability of foreign vendors, that do not correspond to the material reality of most of the Brazilian public-school network.

The TIC Educação survey series, conducted by the Regional Center for Studies on the Development of the Information Society (CGI.br et al., 2025), documents persistent connectivity inequalities in Brazilian public schools. Although its 2024 edition reports that 96% of Brazilian schools now have internet access, effective student use remains deeply unequal: while 67% of students in state schools use the internet for pedagogical activities assigned by teachers, in municipal schools that figure falls to only 27%. Institutional connectivity, in other words, does not translate into effective pedagogical access at the student level. A second concern compounds this structural limit, that of student-data sovereignty: routinely transmitting the textual productions of underage students to commercial servers in foreign jurisdictions falls into a gray zone under the Brazilian General Data Protection Law (Brasil, 2018), especially at its interface with the Brazilian Statute for Children and Adolescents in Digital Environments (Brasil, 2025).

Taken together, these two pressures, precarious connectivity and the legal-ethical fragility of sensitive-data traffic, require that any adoption of generative AI in the Brazilian public school treat strictly local, offline inference as a non-negotiable non-functional requirement. The technical window for such an approach was opened by the recent proliferation of Small Language Models (SLMs) in the 1- to 4-billion parameter range, made viable by aggressive quantization (Q4, Q5) and by lightweight *runtimes* able to run inference on CPU on modest machines, without dependence on dedicated graphics acceleration (Abdin et al., 2024; Yang et al., 2024).

Recent work on SLMs concentrates predominantly on general-purpose *benchmarks*, in English, aimed at mathematical reasoning or code generation. It remains open, however, whether open off-the-shelf models, executed locally under simulated public-school infrastructure, are natively capable of operating as automated writing-feedback tutors in Brazilian Portuguese, respecting structured-output contracts and the pedagogical constraints specified in the prompt. In the educational setting that motivates this study, such a model is not an autonomous tutor but a first-pass assistive component whose output a teacher reviews and complements; the benchmark therefore serves to select, among freely available models, the one that best meets these operational criteria. The present study addresses this gap through a controlled diagnostic evaluation of eight off-the-shelf SLMs against a rubric derived from the Textual Linguistics of Ingedore Koch (Koch, 2018; Koch, 2020), in a rigorously local inference environment and without any external grammatical coercion mechanism.

This work is guided by two related but distinct research questions, one structural-quantitative and the other pedagogical-qualitative:

- **RQ1.** To what extent do contemporary open SLMs, executed locally in an offline environment, adhere to the structural output contract declared in natural language?
- **RQ2.** Among the structurally conformant models, to what extent do their responses mobilize Koch's metalinguistic rubric beyond the thematic-surface layer?

The article is organized as follows: Section 2 reviews the relevant state of the art; Section 3 details the experimental methodology adopted, with a level of description sufficient for integral reproducibility of the study; Section 4 reports computational efficiency; Section 5 presents structural conformance, addressing **RQ1**; Section 6 conducts a qualitative analysis of pedagogical adherence in the best-classified model, addressing **RQ2**; Section 7 introduces a specialist human reference and describes the repertoire of mediation functions that grounds the pedagogical construct; and Section 9 synthesizes conclusions and outlines the subsequent agenda, returning explicitly to each research question.

2 Related Work

Systematic evaluation of language models for educational tasks articulates across three research fronts relevant to the present study. The first front comprises studies on the use of proprietary LLMs as tutors and formative evaluators. Kasneci et al. (2023) synthesize opportunities and challenges of ChatGPT in basic education, identifying significant potential but warning against systemic biases and infrastructural dependence. Tlili et al. (2023) conduct a longitudinal case study on the pedagogical use of *chatbots*, highlighting tensions between student engagement and conceptual fidelity. This literature, however, largely assumes continuous connectivity,

a gap the present work addresses directly.

The second front corresponds to technical literature on SLMs viable for local execution. Abdin et al. (2024), describing Phi-3, posit the thesis of “small and capable models” through intensive training-data curation; Yang et al. (2024) present the Qwen 2.5 family, with emphasis on multilingualism; Grattafiori et al. (2024) and the Gemma team (Gemma Team, 2024) describe, respectively, the Llama 3 and Gemma 2 families in their lightweight variants. These references make up the candidate set of the present *benchmark*, but their technical reports assess performance mostly on Anglocentric *benchmarks* and mathematical tasks, with no specific coverage of pedagogical tutoring in Brazilian Portuguese.

The third front involves the emerging literature on **format bias** in models post-trained by preference. Zheng et al. (2023) and Wu & Aji (2023) demonstrate that human and model-judge evaluators exhibit systematic preference for long, listed, and well-formatted responses regardless of content quality, and that this reward signal becomes internalized in finalized models as a dominant attractor in the generation space. This theoretical body is mobilized, in Section 5, for the interpretation of the architectural findings of this study.

The intersection of these three fronts, namely educational application, local viability in Brazilian Portuguese, and post-training failure patterns, remains, to the best of our knowledge, under-explored. The present article helps fill this gap through controlled empirical diagnosis.

3 Methodology

This section describes, with a level of detail directed at integral reproducibility, the experimental protocol adopted. The subsequent subsections document, in order: (i) the local *offline* inference stack that emulates the school computational ceiling; (ii) the corpus of canonical scenarios and the metalinguistic rubric that anchors it; (iii) the sample matrix and HTTP hyperparameters transmitted on each call; (iv) the deterministic structural-conformance validator; (v) the inferential statistical framework applied to categorical comparisons; (vi) the criteria for metalinguistic pedagogical adherence; (vii) canonical examples of inputs and outputs observed in the corpus; (viii) the secondary falsification protocol; and (ix) reproducibility guarantees. The full statistical rationale, the complete inference payload sent to the server, and the source code of the structural validator are reported, respectively, in Appendices A and B, in conformance with the convention that separates scientific argument in the body from implementation documentation in the appendices.

3.1 Computational Environment and Local Offline Inference Stack

All 312 inference calls were conducted in a strictly local and offline environment, with no communication with remote services during the collection phase. This choice deliberately emulates the computational ceiling available in Brazilian public-school computer labs: execution on a Central Processing Unit (CPU), without dedicated Graphics Processing Unit (GPU) acceleration, and with a Random Access Memory (RAM) restriction compatible with the typical installed base of school machines (around 8 GB).

The inference orchestrator was Ollama, a local HTTP server running on the *loopback* interface (`localhost:11434`). Calls were issued as synchronous POST requests to the `/api/chat` endpoint, with a `Content-Type: application/json` header, following the client-server pattern natively exposed by the *runtime*. Models were instantiated in Ollama's default quantization, `Q4_K_M` for most of the architectural families evaluated, a format compatible with the 8 GB RAM limit and widely used in modest-hardware local inference setups. The client layer was written in Python 3.11, using the `requests` library for HTTP transport and the `json` module of the standard library to serialize and deserialize payloads.

3.2 Evaluation Instrument: Scenario Corpus and Metalinguistic Rubric

The evaluation rubric was derived from Koch's taxonomy of textual cohesion (Koch, 2018; Koch, 2020) and her work on the teaching of writing (Koch & Elias, 2016), and was operationalized in thirteen canonical scenarios designed by the authors (Table 1). Each scenario consists of a short text in a school discursive genre (review, summary, retelling, opinion piece, chronicle), attributed to a hypothetical lower-secondary student (8th and 9th grades), that deliberately carries a target cohesive phenomenon for the automated tutor to identify and

problematize. The scenarios are deliberately synthetic. Authentic student essays typically superimpose several deviations at once, which would make it impossible to attribute the model's response to a specific phenomenon; each text is therefore engineered by the authors to isolate a single target phenomenon in controlled, verifiable form, as a diagnostic evaluation requires. The choice of school genres and of canonical literary sources well established in the lower-secondary curriculum, such as the review of Machado de Assis's *A Cartomante* in scenario 1, is equally deliberate: it keeps each scenario ecologically valid for the 8th–9th grade Brazilian classroom, a text that could plausibly occur in that setting even though it is synthetically constructed. The discursive-genre label that opens several scenarios (for example, the *review* in scenario 1) is part of the synthetic stimulus, not a manipulated variable: genre is not crossed with the target phenomenon in the sample matrix, and the labels serve only to anchor each text in a plausible classroom task. Genre prototypicality was subordinated to the isolation of a single target cohesive phenomenon, so a label does not always match the prototypical structure of the named genre (the *review* in scenario 1, for instance, reads closer to a narrative summary). Because genre is not controlled here, we make no claim about its influence on tutor behavior and revisit it as a limitation in Section 8. The taxonomy spans two axes:

- **Referential Cohesion:** ambiguity through inadequate pronominalization; excessive lexical repetition with suppression of ellipsis and substitution by hypernym or pro-form;
- **Sequential Cohesion:** contradictory use of argumentative operators; disorganization of temporal and spatial markers; extreme juxtaposition of clauses without establishment of logical-semantic relations.

The scenarios were designed by the authors on the basis of six works of Brazilian literature well established in the lower-secondary curriculum—Machado de Assis's *A Cartomante* (Machado de Assis, 1896) and *O Alienista* (Machado de Assis, 1882), Fernando Sabino's *O Homem Nu* (Sabino, 1960), Marina Colasanti's *A Mocca Tecel a* (Colasanti, 1982), L'ucia Machado de Almeida's *O Caso da Borboleta At'iria* (Almeida, 1975), and Lima Barreto's *Cemit'erio dos Vivos* (Lima Barreto, 1956)—selected for their potential to foster different reading, interpretation, and writing skills. From these works, synthetic scenarios were developed, each incorporating a predefined textual phenomenon for the purposes of this investigation. Three scenarios (S11–S13) were intentionally written as high-quality texts, serving as positive controls, to investigate how specialists mediate writing that already satisfies the expected textual criteria, shifting the focus from error correction to expansion, reflection, and learner autonomy. Table 1 synthesizes the thirteen scenarios along four dimensions: the literary source, the proposed school task, the deliberately planted textual phenomenon, and the pedagogical objective each scenario is meant to probe.

Table 1. The thirteen canonical scenarios: literary source, proposed school task, deliberately planted textual phenomenon, and main pedagogical objective. Scenarios S11–S13 are positive controls (well-developed texts), designed to probe mediation that shifts from correction to expansion.

#	Literary work	Genre/task	Planted textual phenomenon	Main pedagogical objective
S1	A Cartomante	Book review	Excessive lexical repetition; limited cohesion	Lexical cohesion and referential variation
S2	O Alienista	Summary	Referential ambiguity (multiple uses of “ele”)	Reference resolution and textual clarity
S3	A Cartomante	Opinion piece	Inappropriate argumentative connectives	Logical relations and textual coherence
S4	A Cartomante	Chronicle	Disordered narrative sequence	Temporal organization and narrative progression
S5	O Alienista	Opinion piece	Generic statements without justification	Argument development and elaboration
S6	O Alienista	Summary	Lexical repetition	Lexical variation and cohesion
S7	O Homem Nu	Retelling	Fragmented sentences; limited cohesion	Sentence integration and narrative fluency
S8	A Mocca Tecel a	Commentary	Limited interpretation; pronominal repetition	Interpretation and textual cohesion
S9	O Caso da Borboleta At'iria	Personal response	Fragmented listing of events without development	Narrative expansion and textual progression

#	Literary work	Genre/task	Planted textual phenomenon	Main pedagogical objective
S10	Cemit'erio dos Vivos	Book review	Contradictory use of connectives	Argumentative coherence and logical consistency
S11	A Cartomante	Book review	Well-developed text (positive control)	Expansion rather than correction
S12	O Alienista	Summary	Well-developed text (positive control)	Reflection and deeper analysis
S13	O Homem Nu	Retelling	Well-developed text (positive control)	Enrichment rather than correction

Each model received a canonical *system prompt*, identical across all experiments to keep this variable under strict control, that defines: (i) the model's role as a writing-feedback tutor for 8th–9th grade students in the Brazilian public school; (ii) five non-negotiable pedagogical constraints (do not provide a ready answer; do not rewrite the corrected text; value linguistic varieties and orality marks without acting as a “grammar police”; focus on cohesion and coherence in Koch's sense; bridge with Computational Thinking when appropriate); and (iii) an output format specified strictly as a JSON object with exactly two keys, `pontos_fortes` (a monolithic string) and `perguntas_reflexivas` (an array of string, with a pair of elements exemplified in the canonical schema). The keys are kept verbatim in Brazilian Portuguese because they are the literal contract sent on every API call.

3.3 Experimental Protocol: Sample Matrix and Inference Hyperparameters

The sample matrix comprised eight distinct language models, distributed across two ranges:

- **Deployable range (analytical center of the study):** `qwen2.5:1.5b-instruct`, `qwen2.5:3b-instruct`, `llama3.2:1b`, `llama3.2:3b`, `gemma2:2b`, and `phi3:mini (3.8 B)`. The delimitation at up to 3.8 billion parameters corresponds to the memory envelope executable in Q4 quantization within the 8 GB RAM target.
- **Reference ceiling (upper baseline):** `llama3:8b` and `gemma2:9b`. These models are not deployment candidates, included as comparative references of attainable quality by the same architectural family at larger scales.

Each model × scenario pair was submitted to three independent repetitions ($n=3$), for a total of $8 \times 13 \times 3 = 312$ inference calls. Triple replication lets us estimate the standard deviation and the intra-model coefficient of variation ($CV = \sigma/\mu$), the metrics needed to separate expected stochastic variability, which derives from the sampling parameter `temperature`, from systemic instruction instability.

The HTTP payload sent to the server was rigorously uniform across all calls, transporting the canonical *system prompt*, the scenario text in the `user` slot, and an options block configuring `temperature = 0.2`, `stream = false`, `format = "json"`, and a client-side `timeout = 120` seconds. The choice of `temperature = 0.2` establishes a compromise between contract adherence and the micro-variability necessary for meaningful CV estimation across repetitions; `stream = false` ensures a single, well-defined termination point for *wall-clock* latency measurement; and `format = "json"` activates Ollama's native syntactic nudging toward parseable JSON output without enforcing any field-typology or key-cardinality schema. This last choice was deliberate, aimed at gauging each model's native adherence to the structural contract specified in natural language, under conditions representative of realistic school deployment. The complete payload and the per-hyperparameter rationale appear in Appendix B.

3.4 Structural Validation: Deterministic Conformance Criteria

Each response was classified for conformance by a deterministic validator that decides, from the raw content returned by the model, whether the response fully satisfies the contract declared in the *system prompt*. The validator applies four disjoint criteria in sequence, and a violation of any one is enough to classify the response as divergent. **C1 (JSON syntax):** the returned string must be a valid JSON document under the ECMA-404 specification, confirming that the decoder respected the syntactic regime induced by Ollama's `format = "json"` parameter. **C2 (root is an object):** the outer structure must be a JSON object (a nominal *mapping*),

since the declared contract prescribes two named keys; the typical violation here is returning an *array* at the root, a pattern inherited from tabular output regimes and incompatible with an individual tutoring interface. **C3 (pontos_fortes is a non-empty monolithic string):** the canonical schema exemplifies this field as continuous prose, oriented toward a welcoming register through the recognition of textual merits; replacing that prose with an enumeration shifts the register from welcoming feedback to corrective evaluation, in direct violation of the “do not act as grammar police” directive. **C4 (perguntas_reflexivas is a non-empty list of strings):** the contract explicitly declares the array of `string` type with a minimum of two items; the validator adopts the more permissive criterion of cardinality ≥ 1 , and divergences at this level almost always appear as types other than `string` (nested objects or `null` values).

The choice for a deterministic classifier, rather than evaluation by a judge model, was deliberate: structural conformance is a syntactic-typological property decidable in finite time by type inspection, with no room for interpretive subjectivity, and delegation to an LLM judge would introduce undesired variance precisely on the dimension where the study seeks maximal objectivity. The full source code of the `divergente()` function, in the exact order of execution applied over the 312-record corpus, is reported in Appendix B.

3.5 Inferential Statistical Framework

Categorical comparisons that support the structural-conformance claims of this paper rely on Fisher's exact test on 2×2 contingency tables, complemented by Wilson 95% confidence intervals for each observed proportion. The full statistical rationale, including the four converging reasons for choosing Fisher's exact test, the treatment of the odds ratio under zero-cell contingencies, and the APA-style significance thresholds, is given in Appendix A. Latency data, by contrast, are reported only descriptively (Section 4), since the separation between model classes is evident at the order-of-magnitude scale and needs no inferential support.

3.6 Analysis Criteria: Efficiency and Metalinguistic Pedagogical Adherence

Computational efficiency. Latency was measured in *wall-clock* via `time.time()` in Python, including the entire local I/O stack between client and Ollama server. For each model, the following were computed: mean, standard deviation, median (p50), 95th percentile (p95), minimum, and maximum, in milliseconds. The coefficient of variation (CV) operates as a dimensionless measure of stability across repetitions. Input and output token counters were obtained from Ollama's native telemetry (`prompt_eval_count` and `eval_count`), preserving fidelity of the measure relative to each model's specific tokenizer.

Metalinguistic pedagogical adherence. For the model best ranked on efficiency and structural conformance, we carried out an additional qualitative inspection, checking whether the response text contains terms from Koch's metalinguistic taxonomy: cohesion, connector, pronoun, referential, sequential, ambiguity, repetition, synonym, ellipsis, argumentative operator, temporal marker, juxtaposition. The presence of at least one rubric term was taken as a minimum indication of metalinguistic adherence; its absence classifies the response as operating in an exclusively thematic-surface layer. Unlike the deterministic structural validator of Subsection 3.4, this lexical criterion is a coarse lower-bound screen, not a decidable property: the presence of a term does not by itself guarantee genuine metalinguistic scaffolding, nor does its absence strictly preclude it. We therefore treat the resulting rate as an indicative triage measure rather than a validated construct, and return to its validity limits, including the planned validation against specialist human judgment via Cohen's κ , in Section 8.

3.7 Anatomy of Outputs: Canonical Examples from the Corpus

To expose the nature of the contracts evaluated, this subsection illustrates, with literal material extracted from the corpus, the cohesive deviation embedded in a representative scenario, one conformant output, and two distinct modalities of divergent output.

Input (scenario 1, Referential Cohesion: excessive lexical repetition). Synthetic text attributed to a hypothetical student, taken literally from the canonical thirteen-scenario corpus (kept in Brazilian Portuguese as the object of linguistic study):

Resenha sobre A Cartomante: A cartomante falou pro Camilo que tava tudo bem. A cartomante leu as cartas. A cartomante sorriu pra ele. A cartomante deu certeza pra ele. Depois o Camilo foi pra casa do Vilela e morreu. A cartomante tinha errado tudo.

The embedded target phenomenon is the lexical repetition of the noun phrase “a cartomante” across four consecutive clauses, a clear case of referential cohesion through repetition. Here the tutor would be expected to nudge the student toward substitution by a pronominal pro-form or a hypernym, prompting reflection on the referential chain without handing over the corrected form.

Translation note for non-Lusophone readers (not part of the experiment): “Review of ‘The Fortune-Teller’. The fortune-teller told Camilo everything was fine. The fortune-teller read the cards. The fortune-teller smiled at him. The fortune-teller gave him certainty. Then Camilo went to Vilela’s house and died. The fortune-teller had got it all wrong.”

Conformant output (qwen2.5:3b-instruct, repetition 1). The model strictly respects the typology declared in the *system prompt*: `pontos_fortes` is returned as continuous prose and `perguntas_reflexivas` as a pair of questions in a list of *strings*.

```
{
  "pontos_fortes": "O texto tem uma estrutura boa, com um inicio claro e uma conclusao que ressalta a importancia d
  "perguntas_reflexivas": [
    "Como voce imagina o que aconteceria se a cartomante tivesse acertado tudo?",
    "Por que e importante ler entre as linhas nos livros?"
  ]
}
```

Structural adherence does not entail pedagogical adequacy: the conformant response above shifts to the thematic plane of the literary work (the importance of reading), without mobilizing any metalinguistic terminology, a point taken up qualitatively in Section 6.

Divergent output, modality A (llama3.2:1b, repetition 1, typological violation of `pontos_fortes`). The model emits syntactically valid, parseable JSON with both expected keys; the divergence appears only as the replacement of the monolithic prose by an itemized enumeration, in direct violation of C3 of the `divergente()` algorithm:

```
{
  "pontos_fortes": [
    "A Cartomante falou pro Camilo que tava tudo bem.",
    "A Cartomante leu as cartas.",
    "A Cartomante sorriu pra ele.",
    "A Cartomante deu certeza pra ele."
  ],
  "perguntas_reflexivas": [
    "Ela falou, mas nao disse nada.",
    "O que a Cartomante fez?",
    "Por que a Cartomante sorriu?"
  ]
}
```

Divergent output, modality B (phi3:mini, repetition 1, full schema substitution and pedagogical inversion). The model abandons the declared canonical schema and imposes a structure of its own, built around nested objects with unsolicited keys (`erro`, `correção`, `texto_original`, `texto_proposto`), violating at once C3 (type of `pontos_fortes`) and the anti-prescriptivist pedagogical directive of the *system prompt*. Illustrative excerpt:

```

{
  "pontos_fortes": [
    {
      "erro": "A palavra 'tava' e uma forma coloquial da conjugacao do verbo estar, que pode ser considerada inform
      "correcao": {
        "texto_original": "A cartomante falou pro Camilo que tava tudo bem.",
        "texto_proposto": "A cartomante falou para Camilo que estava tudo bem."
      }
    },
    {
      "erro": "O uso de 'leu' e um erro comum na conjugacao do verbo ler no passado, pois a forma correta seria 'le
      "correcao": { "texto_original": "A cartomante leu as cartas.", "...": "..." }
    }
  ]
}

```

Note the double anomaly: (i) the model classifies the legitimate colloquialism “tava” as an error to be corrected, in violation of the explicit directive to value linguistic varieties; and (ii) it hallucinates a non-existent grammatical rule by prescribing the form “lera” (pluperfect indicative) in place of “leu” (simple past), the form correctly used by the student. In doing so it confuses distinct verbal aspects and introduces an error into the original text under the pretext of correction. This case confirms the multidimensional collapse pattern discussed in Subsection 5.2.

3.8 Secondary Falsification Protocol via Counter-Experiment

Beyond the main protocol described above, we designed and executed a secondary falsification protocol aimed specifically at the study's most salient architectural finding, the systematic divergence observed in the Llama 3.2 family (1B and 3B). The motivation was twofold: (a) to rule out competing hypotheses that would reduce the finding to an instrumental artifact, such as configuration error, stochastic instability, or interference from the Python client layer; and (b) to characterize the precise nature of the bias, distinguishing *compositional incapacity* from *disobedience under a zero-shot regime*. The protocol was operationalized in four disjoint isolations, run on the same canonical scenarios as the main protocol and under the same local Ollama inference stack:

1. **E1 (stochasticity isolation).** 26 calls (2 models × 13 scenarios × 1 repetition) with `temperature = 0.0`, a deterministic regime in which the decoder uses strict greedy sampling. If the failure persists here, sampling variability is ruled out as a cause.
2. **E2 (in-context demonstration isolation).** 24 calls (2 models × 12 scenarios × 1 repetition), each preceded in the message sequence by a `user/assistant` pair showing scenario 1 and its canonical gold response in strict schematic format. Scenario 1 is consumed as the demonstration; the remaining twelve are evaluated under minimal contextual anchoring (*one-shot in-context learning*). If divergence reverses here, the model's compositional capacity is intact and the bias is an obedience phenomenon under the zero-shot regime.
3. **E2b (joint isolation).** A combination of E1 and E2: in-context demonstration under `temperature = 0.0`, across 24 calls, to check whether the reversal seen in E2 holds once every source of stochastic variability is suppressed.
4. **E3 (client-layer isolation).** The base calls repeated through the `curl` command-line utility, sending the same JSON payload directly to the Ollama endpoint with no mediation from the Python `requests` library. If the failure persists here, interference from the client transport layer is ruled out.

The two primary isolations (E1 and E2) act on theoretically independent axes, sampling temperature and contextual anchoring; together with E2b and the instrumental isolation E3 they add 80 calls, under the same canonical `SYSTEM_PROMPT`, the same local Ollama, and the same scenarios of the main corpus. Results are reported in Subsection 5.3.

3.9 Reproducibility and Artifact Availability

The complete corpus of the 312 calls, including the *prompts* sent, the raw responses in `resposta_ia`, the telemetry metrics (`tokens_input`, `tokens_output`, `latencia_ms`), and the execution metadata

(`modelo_testado`, `cenario_id`, `rep`), was preserved in two structured JSON files of 312 homogeneous records, in long-matrix format. The collection, deterministic classification, and aggregation scripts were written in Python 3.11, using only the standard library and the packages `requests`, `pandas`, `matplotlib`, and `scipy`. The full source code and raw data are publicly available in the authors' repository, following the Open Science practices recommended by the Brazilian Computer Society.

4 Computational Efficiency

Table 2 synthesizes the aggregated efficiency metrics per model (n=39 calls per row), ordered by increasing mean latency.

Table 2. Efficiency metrics per model.

Model	Mean latency (ms)	Std. dev. (ms)	p95 (ms)	CV	Output tokens (mean)
qwen2.5:1.5b-instruct	2,299	474	3,234	20.6%	118.8
llama3.2:1b	2,389	1,423	6,391	59.6%	119.5
qwen2.5:3b-instruct	2,906	552	4,486	19.0%	104.6
llama3.2:3b	3,020	695	4,721	23.0%	108.9
gemma2:2b	3,744	720	5,080	19.2%	121.8
gemma2:9b	8,681	1,425	11,130	16.4%	126.9
llama3:8b	10,523	19,708	19,326	187.3%	144.7
phi3:mini	14,769	12,228	43,717	82.8%	471.2

The classical Human–Computer Interaction literature establishes three system-response thresholds relevant to user experience: up to 100 ms for the sensation of instantaneity; up to 1 s for the preservation of cognitive flow; and up to 10 s for keeping the user's directed attention in sequential tasks (Nielsen, 1993; Card et al., 1991). This last threshold is the operational criterion we adopt for pedagogical viability in synchronous classroom use: above it, the student disengages from the task while waiting for the tutor's response, breaking the continuity of metalinguistic reflection.

Against this criterion, the Qwen 2.5 family leads clearly on efficiency, with mean latencies of 2.299 s (1.5 B variant) and 2.906 s (3 B variant). Both models sit at less than a third of the UX threshold, and their coefficients of variation ($CV \approx 19\text{--}21\%$) indicate that the observed variability is predominantly stochastic, attributable to the `temperature = 0.2` parameter, rather than to systemic instability. In both cases the 95th percentile of latency stays below 4.5 seconds, so that even executions at the upper end of the distribution do not break the time contract with the user.

By contrast, `phi3:mini` is a clear case of computational infeasibility for the synchronous educational scenario. Its mean latency of 14.769 s exceeds the HCI threshold by more than 47%, and the distribution shows that the problem is structural rather than incidental: a standard deviation of 12,228 ms (CV of 82.8%) and a p95 of 43,717 ms reveal markedly bimodal behavior, with prolonged-generation episodes that can exceed three quarters of a minute in extreme cases. This temporal instability is compounded by a mean output verbosity of 471.2 tokens, between 3.9 and 4.5 times that of the other models, which range from 104 to 145 tokens. Such verbosity is itself a pedagogical failure: effective tutorial feedback is brief and focal rather than an extended exposition, and the textual inflation seen in Phi-3 signals a misalignment between the model's generation regime and the communicational demands of the task.

We also note the anomalous behavior of `llama3:8b` at the upper end of the distribution, with a CV of 187.3% and an absolute maximum of roughly 130 seconds in an isolated run. Although it lies in the study's comparative-ceiling range and is therefore not a deployment candidate, this finding recommends caution when adopting 7–8 billion parameter models in comparative research that relies on temporal metrics.

5 Structural Conformance and Architectural Findings

Conformance to the structural contract specified in natural language, namely a JSON object with the keys `pontos_fortes` (type `string`) and `perguntas_reflexivas` (type `array` of `string`), yields the most striking finding of this study. Table 3 consolidates the observed conformance rates, with Wilson 95% confidence intervals and the two-sided Fisher exact p -value for each model relative to `qwen2.5:3b-instruct` as the reference.

Table 3. Structural conformance per model, with Wilson 95% CI and Fisher exact test against the reference model.

Model	k/n	Conf. rate	Wilson 95% CI	Tier	Fisher vs. Qwen 3B
<code>qwen2.5:3b-instruct</code>	39/39	100.0%	[91.0%; 100.0%]	deployable	reference
<code>gemma2:9b</code>	39/39	100.0%	[91.0%; 100.0%]	ceiling	$p=1.000$ (n.s.)
<code>llama3:8b</code>	39/39	100.0%	[91.0%; 100.0%]	ceiling	$p=1.000$ (n.s.)
<code>qwen2.5:1.5b-instruct</code>	38/39	97.4%	[86.8%; 99.5%]	deployable	$p=1.000$ (n.s.)
<code>gemma2:2b</code>	38/39	97.4%	[86.8%; 99.5%]	deployable	$p=1.000$ (n.s.)
<code>llama3.2:1b</code>	0/39	0.0%	[0.0%; 9.0%]	deployable	*** $p<0.001$
<code>llama3.2:3b</code>	0/39	0.0%	[0.0%; 9.0%]	deployable	*** $p<0.001$
<code>phi3:mini</code>	0/39	0.0%	[0.0%; 9.0%]	deployable	*** $p<0.001$

Note. k= conformant responses; n=39 calls per model. *** $p<0.001$, two-sided Fisher exact, $\alpha=0.05$. When a cell of the contingency table is zero, the odds ratio is not finitely estimable; the reported p-value derives from the exact hypergeometric distribution with continuity correction. Wilson 95% CIs computed without normal approximation.

The distribution is clearly bimodal: five models show integral or near-integral adherence (97.4–100.0%, Wilson 95% CI: [86.8%; 100.0%]), while three collapse entirely (0.0%, Wilson 95% CI: [0.0%; 9.0%]), with no intermediate cases. Fisher's exact test confirms that the conformance rates of the Llama 3.2 family (both 1B and 3B variants) and of Phi-3 Mini differ from the reference model at $p<0.001$ each, while the four other conformant models, including the 7–9 B reference-ceiling pair, are statistically indistinguishable from the reference ($p=1.000$ n.s.). This last point matters methodologically: the deployable model `qwen2.5:3b-instruct` is not detectably inferior in structural conformance to the larger reference-ceiling models, which strengthens its choice as the operationally preferable base. The polarization suggests that conformance to the contract is not a monotonic function of parameter count, but an architectural-behavioral property of the family and of its instruction-tuning regime.

5.1 The Family Failure of the Llama 3.2 Line

Qualitative inspection of the two Llama 3.2 variants (1 B and 3 B) reveals an essentially identical failure pattern: across all 78 calls the emitted JSON is syntactically valid and parseable, and the divergence appears only as a typological violation of the `pontos_fortes` field, returned consistently as an *array* of *strings* (typically two to four items) in place of the specified monolithic `string` type. This perfect symmetry between the two sizes, alongside the full conformance of the Qwen 2.5 family at comparable sizes under an identical *prompt*, rules out insufficient capacity or stochasticity and points the explanation toward post-training.

As reported in Table 3, the pairwise comparison of the Llama 3.2 family against the reference model `qwen2.5:3b-instruct` yields $p<0.001$ for both variants (1B and 3B), while the matched-scale Qwen 2.5 family is statistically indistinguishable from the reference ($p=1.000$ n.s.). The cross-family separation, at matched scale and under identical natural-language specification, is therefore not a borderline result subject to interpretation: it is one of the most categorical statistical contrasts observable in 2×2 categorical experimentation. As noted in the table caption, the odds ratio is not finitely estimable when one cell of the contingency table is zero; the exact p -value alone, derived from the hypergeometric distribution, suffices to falsify the null hypothesis of homogeneous conformance rates between families at any conventional significance threshold.

This pattern is coherently interpreted in light of the literature on format bias in preference-post-trained models. Zheng et al. (2023) and Wu & Aji (2023) document that implicit format-based rewards, whether operating

through human evaluators or judge models, induce a systematic inflation of stylistic markers such as lists, *bullet points*, bold headings, and enumerations, at the expense of *prompt* fidelity. The motivation behind this bias is direct: responses structured as marked items are perceived as more professional and readable by human evaluators in general-purpose assistant interfaces, and the reward signal internalizes this preference as a dominant attractor in the generation space.

Liu et al. (2025), in the GuideEval framework, provide the behavioral-theoretical complement needed to interpret the finding: the framework decomposes a model's tutorial competence into three phases, the Perception of the student's cognitive state, the Orchestration of the tutorial response, and the Elicitation of reflection, and shows that failures in the Orchestration phase typically appear as rigidity toward format instruction, with the model retreating to output patterns strongly entrenched in post-training. In these terms, the anomaly observed in the Llama 3.2 line is precisely an Orchestration failure: the model *perceives* the task correctly (assisting the student) and *would elicit* questions, but the intermediate stage of behavioral modulation collapses to the enumeration attractor conditioned by its generalist post-training. The present study adds two points to this literature. First, the formatting bias is resistant to explicit counter-specification in natural language under a zero-shot regime: the *system prompt* declares the expected type unambiguously, yet the enumeration attractor prevails in 100% of the calls. Second, the bias is an architectural signature consistent across scales of the same family, which suggests that the phenomenon originates in the institutional post-training procedure adopted by the team responsible for the 3.2 line rather than in the particularities of an isolated *checkpoint*. The operational implication, however, needs the qualification provided in Subsection 5.3: in the absence of external syntactic coercion and under a pure zero-shot regime, the standard condition of most economically viable educational deployments, the Llama 3.2 family is not deployable as structured-tutoring infrastructure in Brazilian Portuguese; under minimal contextual anchoring by demonstration (*one-shot*), however, conformance is fully recovered, which reclassifies the finding as a reversible zero-shot bias rather than a categorical compositional incapacity of the model.

5.2 The Multidimensional Collapse of Phi-3 Mini

The failure of `phi3:mini` is qualitatively distinct and more severe. The model not only typologically violates the `pontos_fortes` field, emitted as an *array* in 100% of calls, but replaces the declared schema entirely with one of its own, built around the unsolicited fields `erro`, `correção`, `texto_original`, and `texto_proposto`. In addition, 20.5% of calls present `perguntas_reflexivas` as `NoneType` or composed of nested non-*string* elements, a progressive deterioration of contract fidelity as generation advances. The Fisher exact comparison of Phi-3 Mini against the reference model (Table 3) yields $p < 0.001$, of the same magnitude as the Llama 3.2 contrasts, which justifies bracketing it analytically in the same regime of zero-shot non-adherence.

More serious, however, is the pedagogical inversion of the *prompt*. Trained predominantly on English-language corpora for general-purpose assistant use, the model adopts a corrective-prescriptivist posture in flagrant violation of the explicit directive of the *system prompt*, “value linguistic varieties and orality marks”. It classifies, for instance, the legitimate colloquialism “*tava*” as an error to be corrected to “*estava*” and then hallucinates a non-existent grammatical rule by prescribing the form “*lera*” (pluperfect) in place of “*leu*” (simple past), the form correctly used by the student.

This double violation, structural and pedagogical, confirms the general hypothesis of Wu & Aji (2023) on the imperfect transferability of instruction habits across distinct linguistic-pragmatic domains. Optimized for an Anglocentric conversational regime in which explicit correction is virtuous, the model carries this habit into a pedagogical-linguistic context in Brazilian Portuguese where it is harmful, and it does so even under counter-instruction in natural language. The combination seen in Phi-3, structural collapse, pedagogical inversion, temporal instability, and quadruple verbosity relative to its peers, is a clear case of the transferability limit of Anglocentric SLMs to educational tasks in Brazilian Portuguese, even when the developers nominally declare the model multilingual.

Phi-3's inflated verbosity, 471 mean tokens against the 104–145 range of the other models, also echoes MathTutorBench by Macina et al. (2025), which systematizes, in an adjacent domain (mathematical tutoring), the finding that off-the-shelf models tend to superimpose thematic expertise over pedagogical competence, turning the tutorial act into an extended expository monologue rather than brief, focused guidance. The verbal excess observed here is therefore not a quirk of tokenization or instance noise: it is a local manifestation of a systemic pattern in post-trained generalist models, which conflate output volume with tutorial quality, to the direct detriment of the focal, dialogic regime that effective tutorial mediation requires.

5.3 Robustness Analysis of the Llama 3.2 Architectural Finding via Controlled Counter-Experiment

The perfect uniformity of divergence in the Llama 3.2 family (78 calls, two architectural sizes, and zero conformant responses in the main protocol) demands, by sound scientific practice, a deliberate examination of competing hypotheses that would reduce the finding to an instrumental artifact before it can be sustained as a behavioral signature of the model. To this end, the secondary protocol described in Subsection 3.8 was executed in full, adding 80 calls under the same local inference stack. Table 4 summarizes the outcomes, with Wilson 95% CIs and the two-sided Fisher exact p -value of each isolation against the zero-shot baseline.

Table 4. Structural conformance of the Llama 3.2 family under falsification isolations.

Isolation	Llama 3.2 1B	Llama 3.2 3B	Wilson 95% CI (3B)	Fisher vs. baseline
Baseline (zero-shot, $t=0.2$)	0/39 (0.0%)	0/39 (0.0%)	[0.0%; 9.0%]	—
E1 (zero-shot, $t=0.0$)	0/13 (0.0%)	0/13 (0.0%)	[0.0%; 22.8%]	$p=1.000$ (n.s.)
E2 (one-shot, $t=0.2$)	11/12 (91.7%) ¹	12/12 (100.0%)	[75.8%; 100.0%]	*** $p<0.001$
E2b (one-shot, $t=0.0$)	9/12 (75.0%) ²	12/12 (100.0%)	[75.8%; 100.0%]	*** $p<0.001$
E3 (curl, zero-shot)	failure reproduced	failure reproduced	—	—

Note. ¹ The single non-conformance of the 1B in E2 derives from HTTP timeout, not from a structural violation of the response actually returned. ² Idem for 3 of 12 of the 1B in E2b: timeouts associated with the larger context budget introduced by the demonstrative one-shot pair, with no bearing on the typological conformance of the completed calls. *** $p<0.001$, two-sided Fisher exact, $\alpha=0.05$. Fisher comparisons reported for the 3B variant against its own zero-shot baseline (0/39); the 1B variant exhibits the same direction of effect under the same conventional thresholds.

The joint reading of the four isolations sustains three interlinked conclusions, of falsificationist nature, on the nature of the finding.

First conclusion: the bias is deterministic, not stochastic. In E1, suppressing all sampling variability with `temperature = 0.0` produces no conformance; on the contrary, the 3B model converges to the structural attractor `pontos_fortes = list[str], cardinality = 2` in 12 of the 13 scenarios, the near-degeneration profile typical of a bias encoded in the weight space. Hypotheses that would attribute the divergence to sampling noise or to instability across repetitions are therefore falsified.

Second conclusion: the bias lies in the model, not in the client stack. In E3, sending the same JSON payload to the Ollama endpoint via `curl`, with no mediation from the Python `requests` library, reproduces the typological-divergence pattern of the main protocol in full: `pontos_fortes` returned as a list, JSON syntactically valid, the other keys consistent. Hypotheses that would attribute the divergence to interference from the client transport layer, the HTTP headers, the payload encoding, or the serialization of the `messages` array are therefore falsified.

Third conclusion: the bias is one of zero-shot obedience, not of compositional incapacity. In E2, placing a single `user/assistant` pair with the canonical gold response for scenario 1 ahead of the query reverses the divergence completely: the 3B model reaches 12/12 conformance in the twelve following scenarios, and the 1B model reaches 11/12 (the single exception being a transport *timeout*, not a format failure). This reversal is stable in E2b, where combining *one-shot* with `temperature = 0.0` keeps 100% conformance in the 3B. Fisher's exact test on the 2×2 condition-by-conformance contingency (Table 4) corroborates it: for the Llama 3.2 3B

variant (12/12 conformant under one-shot vs. 0/39 baseline) and for the 1B variant (11/12 conformant, the single non-conformance again due to an HTTP timeout rather than a structural violation), the contrast yields $p < 0.001$. The determinism isolation E1 (0/13 vs. 0/39 baseline), by contrast, yields $p = 1.000$ n.s., confirming that the bias is invariant to sampling temperature. The inference is direct: the Llama 3.2 models do possess the compositional capacity to produce `pontos_fortes` as a monolithic string that respects the declared schema; what fails is obedience to that specification when it is given only in natural language, under a zero-shot regime, without a contextual demonstration. The enumeration attractor dominates only when no contrary demonstrative anchor is present.

This set of conclusions repositions the finding of Subsection 5.1 with greater precision. The Llama 3.2 family does not show a categorical incapacity to operate as a structured tutor; it shows a reversible zero-shot bias, in the strict sense of Liu et al. (2025): a failure in the Orchestration phase of the tutorial response when the operating regime provides no demonstrative reference, with full recovery once such a reference is given. This characterization has three implications:

1. **Methodological implication for the paper.** The choice of `qwen2.5:3b-instruct` as the base model for subsequent pedagogical fine-tuning remains justified: among the models evaluated, it is the only one conformant under zero-shot, and zero-shot is the regime of lowest engineering cost in the classroom. The justification, however, is no longer “Llama 3.2 is not deployable” but “Qwen 2.5 3B is the only model that needs no additional prompting engineering to guarantee the structural contract”.
2. **Operational implication for alternative deployments.** This establishes a valid intermediate route for schools or municipal networks that, for pedagogical or technical-sovereignty reasons, prefer the Llama 3.2 family: the additional engineering cost is keeping a 300-token demonstrative pair in the context on each call, a moderate increase in budget, though with an observable latency penalty, as the *timeouts* in E2/E2b over the 1B show.
3. **Implication for the literature on format bias.** The present study provides controlled experimental evidence that the bias documented by Zheng et al. (2023), Wu & Aji (2023), and Liu et al. (2025) is not absolute: it depends on the operating regime adopted. This refinement, obtained from a directed falsification protocol, gives a basis for future work on the computational economy of educational *prompt engineering* as a way to mitigate post-training bias in off-the-shelf SLMs.

A further mitigation route, orthogonal to the one-shot anchoring examined above, deserves explicit mention so that the structural finding is not over-read. The same Ollama runtime used here also supports schema-constrained decoding, in which a JSON Schema is compiled into a grammar that masks structurally invalid tokens at each decoding step, turning contract conformance from a probabilistic nudge into a hard guarantee. Under such a regime the structural failures reported in Table 3 would very likely disappear for every family, as the literature on grammar-constrained generation documents for sub-7 B models (willardlout2023, ; dong2024xgrammar,). We deliberately excluded this mechanism from the main protocol because the object of study is precisely native adherence to a contract stated in natural language: schema coercion would mask the very post-training bias the study sets out to measure. The structural finding should therefore be read as a property of the zero-shot, no-coercion deployment regime, the lowest-engineering-cost classroom configuration, and not as a claim that these models are incapable of emitting valid structure under any configuration.

6 Qualitative Analysis: Pedagogical Adherence of the Conformant Model

Taken together, Sections 4 and 5 single out `qwen2.5:3b-instruct` as the one configuration among the *deployable* models that satisfies all non-functional requirements at once: a mean latency of 2.906 s (below the HCI threshold), controlled variability (CV = 19.0%), and full adherence (0/39) to the JSON contract without recourse to external coercion. This result, however, does not license an optimistic inference about the model's pedagogical adequacy, as the additional qualitative inspection below shows. We flag a deliberate change in evidentiary regime here: whereas the structural findings rest on 312 primary and 80 falsification calls with Fisher significance, the analysis that follows is an exploratory single-model probe, and we read it as

corroborating the structural diagnosis rather than as carrying independent inferential weight.

Applying the metalinguistic criterion of Subsection 3.6 to all three repetitions of `qwen2.5:3b-instruct` (n=39 responses), we find that only 20 of 39 (51.3%; Wilson 95% CI: [36.2%; 66.1%]) mobilize any terminology from the cohesion taxonomy. Pooling the three repetitions, rather than inspecting a single one, both narrows the interval and exposes a second and sharper pattern: the mobilization is markedly unstable across repetitions. Of the thirteen scenarios, only one elicits rubric terminology in all three runs, one never elicits it, and the remaining eleven oscillate from run to run, so that the model cannot be relied upon to surface the metalinguistic object even where it is capable of doing so. Table 5 reports the observed proportion together with the cross-repetition stability and the interpretation of the confidence bounds.

Table 5. Metalinguistic adherence of `qwen2.5:3b-instruct` (Wilson 95% CI).

Metric	Result
Responses mobilizing Koch terminology	20/39 (51.3%)
Wilson 95% CI	[36.2%; 66.1%]
Cross-repetition stability (per scenario)	1/13 always, 1/13 never, 11/13 variable
Upper bound (optimistic)	66.1%: absence in $\geq 34\%$
Lower bound (pessimistic)	36.2%: absence in $\geq 64\%$
Classification	Partial and unstable adherence

Note. The width of the CI ([36.2%; 66.1%]) reflects the still-modest sample (13 canonical scenarios over 3 repetitions, n=39) and constitutes an acknowledged limitation of this analysis; nevertheless, even its upper bound implies absence of metalinguistic terminology in at least 34% of cases. The characterization “unstable” is, moreover, not rhetorical: only 1 of the 13 scenarios mobilizes the rubric across all three repetitions, which grounds the instability claim empirically rather than by assertion.

In the runs where no rubric terminology surfaces (close to half of the total), the model shifts the exchange to the thematic-narrative plane of the literary work the student cites (characters, plot, setting), commenting on elements of the literary source rather than evaluating the textuality of the student's production. Its response to the Referential Cohesion scenario in the first repetition, for example, in which the student text repeats the noun phrase “*a cartomante*” across consecutive clauses, merely observes “good structure, clear beginning, and conclusion that highlights the importance of reading”, without any mention of the excessive lexical repetition that was the target phenomenon of the scenario.

This is a categorical task failure: the model conflates *what the student wrote* with *what the student wrote about*, and loses the central pedagogical object of a metalinguistic intervention guided by the rubric. Even when some taxonomy term is evoked, the approach stays largely declarative: the model *names* the linguistic phenomenon without leading the student to *reflect* on how it operates in the text, which contradicts the mediational structure the rubric prescribes.

This finding has direct theoretical support in MathTutorBench by Macina et al. (2025), which documents, in an adjacent educational domain, a systematic trade-off between thematic expertise and pedagogical competence in off-the-shelf models: the same model that can correctly discuss the content of a literary work or a problem tends to fail as a tutor when asked to lead the student through progressive questioning, retreating to an expository-thematic regime. The behavior seen here in `qwen2.5:3b-instruct` is a local manifestation of this pattern: the model is proficient at talking *about* “A Cartomante” as a literary work, offering reasonable commentary on its structure, reading, and plot, but it fails to sustain the metalinguistic-mediational stance that would reposition the focus on the student's text as a linguistic artifact open to problematization in light of Koch's taxonomy. The GuideEval framework of Liu et al. (2025) likewise categorizes this failure mode as a joint deficiency in the Perception phase (the model does not detect the target cohesive phenomenon) and the Elicitation phase (even when it detects the phenomenon, it does not pose questions that make the student operate on the text itself), confirming that format adherence and pedagogical adequacy are logically independent dimensions of a model's tutorial competence.

7 Specialist Human Reference: A Repertoire of Mediation Functions

The qualitative probe of Section 6 reports what the best-classified model fails to do, but it does so against a rubric of linguistic phenomena, not against an articulated account of what an experienced teacher actually does when returning written work to a student. This section supplies that account. It introduces a specialist human reference, collected and analyzed for a single, deliberately bounded purpose: to characterize, in descriptive terms, the repertoire of pedagogical moves that expert teachers mobilize when giving writing feedback to lower-secondary students. We stress at the outset that this reference is *not* used as a performance yardstick against the model. Rather than serving as a gold-standard response set against which the model's outputs are scored, the specialist feedback was analyzed to derive the mediation functions that underpin the pedagogical dimension of the benchmark; the comparison the study cares about is thus model against mediation functions derived from expert practice, not model against teacher. We make no claim of the form “the model is worse than the teachers”; the planted-error design of the scenarios (Subsection 3.2) makes the mere identification of the target phenomenon circular by construction, since the deviation was engineered to be found, so that what a comparison on problem detection would measure is the scenario design rather than either party's competence. The analytical value of the human corpus lies elsewhere: in the *mediational* layer, the strategies by which a teacher formulates the return, decides when to ask rather than tell, how much scaffolding to extend, and whether to model a repair, none of which is dictated by the scenario. We read this layer through the lens of mediation and scaffolding in the Vygotskian tradition (Vygotsky, 1978; Wood et al., 1976), in which feedback is an interactional support calibrated to the learner's zone of proximal development rather than a verdict on a product.

7.1 Panel and Corpus

Five teachers of Portuguese, identified here only by code (E1–E5), each responded to the same thirteen canonical scenarios used in the main benchmark, producing 65 written feedback turns (*devolutivas*) with no missing cells. All participants reported more than ten years of classroom experience; two work precisely in the 8th–9th grade band that the project targets, which anchors the corpus in the instructional reality the eventual platform is meant to support, and one is a nationally recognized reference in Linguistics, which lends weight to the specialist character of the panel. All signed an informed-consent term for anonymous use under a Creative Commons license; the anonymized responses are released with the study's artifacts. Minor typographical slips by the teachers are preserved verbatim and never treated as student language. One cell (E4, scenario 10) is misaligned, its content corresponds to the neighboring scenario 9, and is therefore flagged and isolated from any count organized by target phenomenon, although it remains codable for mediation functions, which do not depend on the scenario's specific content.

7.2 Eight Mediation Functions Emergent from the Corpus

A full reading of the 65 turns yields a recurrent set of eight mediation functions, labeled FM01–FM08, developed within this study and operationalized in a transparent coding manual (codebook v0.2). The functions are: **FM01** recognize competence (naming, specifically, something the student did well); **FM02** name the problem (signaling that some aspect of the text is inadequate, including naming an insufficiency or a genre mismatch, not only an explicit error); **FM03** provoke reflection (an open question that leads the student to re-examine a textual choice without handing over the answer); **FM04** offer a cue (concrete help toward the solution, listing repertoire or indicating the strategy to follow, without rewriting the student's text); **FM05** partially model (demonstrating the repair with a rewritten example sentence); **FM06** propose revision (explicitly requesting the rewriting or correction of what is already in the text); **FM07** challenge expansion (pushing the student beyond repair, to add opinion, analysis, or depth); and **FM08** reinforce autonomy (a closing move that affirms the student's agency and projects continuity). Coding is multi-label and binary at the level of the whole turn: for each function the coder records presence or absence, under a conservative minimal-evidence rule (when in doubt, mark absent), and a single turn typically activates several functions at once.

Table 6 reports the prevalence of each function across the 65 turns. None of the eight is empty: every function emerges from the actual feedback, which supports treating the taxonomy as grounded in expert practice rather

than imposed from theory. Three functions form a near-universal operational core, naming the problem (FM02, 80%), offering a cue (FM04, 69%), and proposing revision (FM06, 57%), while partial modeling (FM05) is rare, occurring in only 9% of turns and used by just two of the five teachers. The mean turn mobilizes 3.8 of the eight functions. Consistent with this, the single most frequent co-occurrence signature is exactly FM02+FM04+FM06 (11 turns), the operational core of naming the problem, pointing the way, and asking for the revision.

Table 6. Prevalence of each mediation function across the 65 feedback turns.

Code	Mediation function	Presence
FM01	Recognize competence	69%
FM02	Name the problem	80%
FM03	Provoke reflection	34%
FM04	Offer a cue	69%
FM05	Partially model	9%
FM06	Propose revision	57%
FM07	Challenge expansion	32%
FM08	Reinforce autonomy	32%
Mean functions per turn	3.8	

7.3 Distinct Mediation Profiles: Variation as the Finding

If the operational core is shared, what *distinguishes* the teachers is not that core but the presence or absence of four other functions. Table 7 reports, for each teacher, the proportion of the thirteen scenarios in which each function appears, together with the mean number of functions per turn, and Figure 1 renders the same profiles visually.

Table 7. Mediation profiles per teacher (% of the 13 scenarios in which each function appears).

Teacher	FM01	FM02	FM03	FM04	FM05	FM06	FM07	FM08	Functions/t urn
E1	100	77	23	69	15	69	0	0	3.5
E2	8	100	62	46	0	62	0	0	2.8
E3	38	69	0	69	0	69	0	0	2.5
E4	100	85	0	85	0	31	69	77	4.5
E5	100	69	85	77	31	54	92	85	5.9

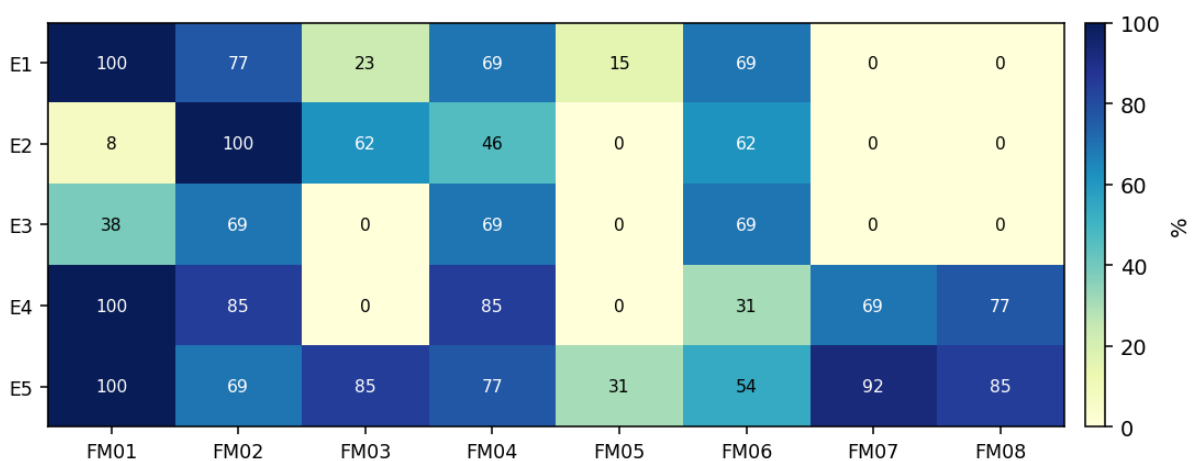


Figure 1. Mediation functions by teacher (% of the 13 scenarios). The operational core (FM02 name, FM04 cue, FM06 revise) is near-universal; the profiles diverge on FM01 (absent in E2), FM03 (only E2 and E5), FM05 (only E1 and E5), and the FM07/FM08 expansion layer (only E4 and E5).

The profiles are coherent and distinct. E3 is telegraphic, restricted to the operational core (FM02/FM04/FM06) with neither reflection, nor modeling, nor an affective-expansive layer, the leanest at 2.5 functions per turn. E2 is austere and reflective without affect: it names the problem in every scenario and often returns it as a question

(FM03 in 62%, e.g. “*Why, all of a sudden, did she leave her husband?*”) backed by a concrete strategy (FM04), yet almost never praises (FM01 in 8%) and never challenges expansion or reinforces autonomy. E1 combines the core with recognition and modeling but no expansion: it recognizes (100%), names, cues, proposes revision, and is one of the two teachers that model a repair (FM05), while never pushing the student beyond the repair (FM07 and FM08 both at 0%). E4 is affective-expansive: strong on recognition (100%), challenge (FM07, 69%), and autonomy (FM08, 77%), but it uses no reflective question (FM03 at 0%) and does not model. E5 is the complete scaffold, high on every function and the only teacher to combine reflection (85%), modeling (31%), and the expansion layer (FM07 92%, FM08 85%), at 5.9 functions per turn. The full mediation repertoire, in short, appears only in the E5 profile; the others are recognizable subsets of it. This is the central observation of the corpus: there is no single way expert feedback is delivered, but a small shared core surrounded by four differentiators, recognition, reflection, modeling, and the expansion-plus-autonomy layer, whose presence or absence defines a teacher's mediational signature.

A recurring boundary deserves a note, because it shapes the coding. When a teacher wants the student to add an opinion or to deepen the text, the *form* of the request decides the function: issued as a directive (“add a brief opinion about Camilo”) it counts as offering a cue (FM04) over a named gap (FM02); posed as an open question (“was the ending surprising?”) it counts as provoking reflection (FM03). The same underlying intent thus separates, on a single well-written text, the directive profile of E4 from the interrogative profile of E5. This directive-versus-question boundary was the main lesson of the reliability round reported next and is fixed in the codebook.

7.4 Inter-Annotator Reliability of the Coding

Because the mediation functions are interpretive, the coding requires an explicit reliability check. A second, blind, and independent annotator (a specialist in Education) coded a stratified sample of 20 turns (4 scenarios × 5 teachers; 160 binary decisions) against the first coding. Table 8 reports Cohen's κ per function. The mean is $\kappa = 0.83$, in the “almost perfect” band of Landis & Koch (1977), and 12 of the 20 turns were coded identically across all eight functions. Reliability is high for the affective and expansive functions (FM01, FM03, FM07, FM08) and lower for the two operational functions FM02 and FM04 ($\kappa = 0.58$ and 0.45), whose boundary with each other and with FM07 is genuinely diffuse.

Table 8. Inter-annotator reliability (Cohen's κ) per mediation function, over a stratified sample of 20 turns.

Code	Mediation function	κ
FM01	Recognize competence	1.00
FM02	Name the problem	0.58
FM03	Provoke reflection	1.00
FM04	Offer a cue	0.45
FM05	Partially model	1.00*
FM06	Propose revision	0.90
FM07	Challenge expansion	0.90
FM08	Reinforce autonomy	0.78
Mean	0.83	

Note. * FM05 was positive in a single case in the sample, so its agreement is trivial and weakly informative. The disagreements were directional rather than random: in 100% of them it was the specialist who marked one *additional* function, never the reverse, concentrated on FM04 (+6) and FM02 (+4). This systematic offset, the first coder under-crediting the naming of an insufficiency and the indication of a path, motivated an inclusive refinement of exactly those two definitions in codebook v0.2 and a recoding of FM02 and FM04 over the full 65 turns. Two caveats follow, stated plainly. First, the reported $\kappa = 0.83$ measures the v0.1 codebook; the inclusive v0.2 recoding of FM02/FM04 was deliberately not re-blinded on the same sample, since doing so would fit the test, so a fresh blind pass on v0.2 is future work rather than a prerequisite for the descriptive reading above. Second, and importantly for the scope of this paper, this κ establishes the reliability of *coding the human repertoire*; it does not validate the lexical keyword screen used for the model in Subsection 3.6. The direct heuristic-versus-human agreement on Koch's terminology, which would validate the RQ2 metric itself,

remains the open item discussed in Section 8.

7.5 Mediation Functions Mobilized by the Conformant Model

The repertoire above was derived from expert practice; we can now turn the same rubric back on the machine and ask, descriptively and without scoring the model against any individual teacher, which of the eight functions the conformant model actually mobilizes. This is the comparison the study cares about, model against the mediation functions derived from practice, rather than model against teacher. We coded the 39 outputs of `qwen2.5:3b-instruct` (13 scenarios $\times$ 3 repetitions) with the same v0.2 codebook and the same minimal-evidence rule applied to the human turns. Two scope caveats frame the reading. First, this is a single-coder pass (the same caveat already noted for the v0.2 human recoding); a blind second pass on the model turns is future work. Second, and more consequentially, the model's output format *pre-loads* part of the repertoire: the `pontos_fortes` field structurally forces a recognition move (FM01) and the `perguntas_reflexivas` field structurally opens space for reflection or expansion. The informative signal, therefore, is not the presence of FM01 but the *absence* of the functions the format does not force, above all the expert operational core of naming the problem, offering a cue, and proposing the revision.

Table 9. Mediation functions mobilized by the conformant model (`qwen2.5:3b-instruct`, $n=39$) against the specialist repertoire ($n=65$). The gap column is in percentage points.

Code	Mediation function	Model	Experts	Gap
FM01	Recognize competence	100%	69%	+31
FM02	Name the problem	3%	80%	-77
FM03	Provoke reflection	10%	34%	-24
FM04	Offer a cue	15%	69%	-54
FM05	Partially model	0%	9%	-9
FM06	Propose revision	0%	57%	-57
FM07	Challenge expansion	41%	32%	+9
FM08	Reinforce autonomy	0%	32%	-32
Mean functions per turn	1.7	3.8		

Table 9 and Figure 2 report the result, and it is sharp. The model's mediation collapses onto two functions: it always recognizes (FM01, 100%, format-forced) and it frequently challenges the student to expand (FM07, 41%), but it almost never enters the corrective core that defines expert practice on a flawed text, naming the problem in 1 of 39 turns (FM02, 3%, against 80% in the human corpus), offering a concrete cue in 15% (against 69%), and proposing a revision in *none* (FM06, 0%, against 57%). It never models a repair (FM05, as is also rare among the teachers) and never closes by reinforcing autonomy (FM08, 0%). The mean turn mobilizes 1.7 functions against the human 3.8, and the few reflective questions it does pose (FM03, 10%) are mostly directed at the plot of the source literary work rather than at the student's own textual choices, which is why the lexical-rubric screen of Subsection 3.6 already flagged its thematic-surface tendency.

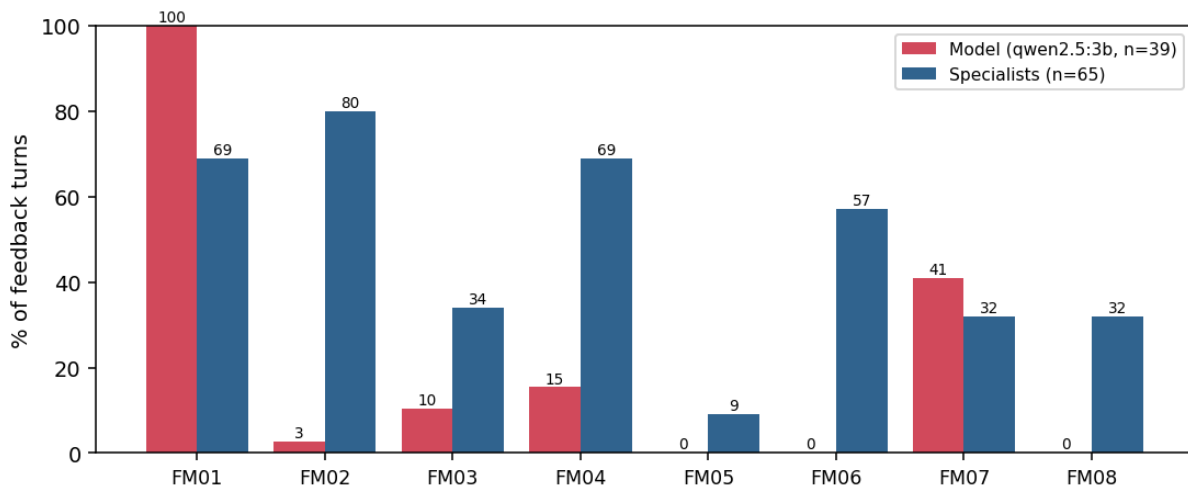


Figure 2. Mediation functions of the conformant model (qwen2.5:3b-instruct, n=39) against the specialist repertoire (n=65). The model is concentrated on the format-forced recognition (FM01) and on open expansion (FM07), and it misses the expert operational core FM02/FM04/FM06.

Two qualitative observations sharpen the diagnosis. First, an active anti-pattern: in 6 of the 30 problematic-text turns the model's recognition move (FM01) *praises as a strength the very phenomenon that was planted as the defect*, commending the “good cohesion” of a deliberately over-repetitive text or the “contradictory connectives” inserted as an error. The model does not merely fail to name the problem; it positively reinforces it. Second, an absence of state-sensitivity: the expert corpus shifts regime with the text, deploying the corrective core (FM02/FM04/FM06) on flawed writing and the expansive layer (FM03/FM07/FM08) on the well-developed positive controls, whereas the model applies the same recognize-and-expand pattern indiscriminately, mobilizing FM07 on flawed and well-formed texts alike and never switching into the corrective regime the flawed scenarios call for. The conformant model thus reproduces only the part of the repertoire that its output schema happens to encode, and leaves precisely the corrective-mediational layer, the one a writing tutor exists to provide, unaddressed.

7.6 Role in the Study and an Open Editorial Question

This specialist reference enters the study on three descriptive fronts, none of which is a performance verdict ranking the model below any individual teacher. First, it supports the *construct validity* of the evaluation: the textual dimensions on which the rubric rests are the same ones around which five independent experts organize their feedback, which lets us state that the rubric is anchored in specialist practice rather than imposed from outside. Second, it provides annotation and reliability material: the 65 turns are real exemplars against which a coding can be calibrated and its agreement estimated, feeding directly the threats-to-validity discussion. Third, and most consequentially, it reframes what counts as a *reference* feedback turn. The variation documented above shows there is no single correct response but a *range* of acceptable mediation profiles that differ in style and in demand; the study therefore describes that range rather than fixing a single gold answer, which turns an apparent weakness, the absence of one canonical target, into a conceptual contribution. The repertoire also defines, in observable terms, the pedagogical target a future specialized model would have to reproduce: which profile the platform should mirror, in complementarity to the teacher rather than in replacement of her, becomes an explicit empirical decision rather than an implicit default. Coding the conformant model against this same rubric (Subsection 7.5) belongs to the same descriptive spirit: it does not rank the model below the teachers but locates, function by function, where the off-the-shelf model falls short of the repertoire, which is precisely what a subsequent specialization would have to target.

This complementarity is worth stating plainly, because it bounds what the benchmark is selecting for. The platform has no pedagogical behavior of its own: the mediation is carried out jointly by the language model, which produces a first-pass return, and by the teacher, who reviews, complements, and contextualizes that return. The mediation functions therefore do not prescribe the platform's behavior; they describe the repertoire of mediations we expect the model to be able to mobilize *in complementarity* to the teacher's work. Read in this

light, the technical and the pedagogical evaluations are not two independent benchmarks but two dimensions of a single one, organized as a chain: an educational problem motivates a repertoire of mediation functions, which supplies the pedagogical criteria against which the technical benchmark of the SLMs is interpreted, which in turn selects, among the deployable models, the one capable of meeting those criteria under the platform's real operating conditions, which finally informs implementation. The mediation functions ground the pedagogical dimension of the benchmark, and the technical evaluation is not an end in itself but is placed at the service of implementing a theoretically grounded pedagogical proposal.

We close with a deliberate note of scope. This section is written to be self-contained, so that the reader, including the co-authors weighing it, can see the technical benchmark and the mediation repertoire as a whole and judge their joint shape. Whether this specialist reference is best kept as a descriptive section of the present diagnostic benchmark, as it stands here, or developed into a dedicated pedagogical study that the technical benchmark would then cite, is an open editorial question that this consolidated version is meant to inform rather than foreclose.

8 Threats to Validity

Five limitations bound the scope of the claims above. **Construct validity of the RQ2 metric.** The metalinguistic-adherence measure of Subsection 3.6 is a lexical keyword screen, a coarse lower-bound proxy rather than a validated instrument: it can miss genuine metalinguistic scaffolding expressed without canonical terminology and can over-credit the mere naming of a phenomenon. Its construct validity should be established by agreement against specialist human judgment (Cohen's κ); the RQ2 figures here are therefore reported as indicative and corroborating, not as a validated measure. We note that a specialist human reference now exists (Section 7), and that the coding of its mediation repertoire reaches an inter-annotator $\kappa = 0.83$; that reliability, however, concerns the descriptive coding of the human turns, not the lexical screen applied to the model, so the direct heuristic-versus-human agreement on Koch's terminology that would validate the RQ2 metric remains the immediate next step for this benchmark. **Corpus size and authorship.** The thirteen scenarios were authored by the research team itself and are deliberately synthetic, a design that buys single-phenomenon attributability at the cost of breadth and of a wide confidence interval on the RQ2 rate; the structural RQ1 contrasts, by contrast, are categorical (0/39 versus 39/39) and do not depend on corpus breadth. **Single-model qualitative scope.** The RQ2 analysis, including both the lexical screen and the coding of the model's outputs against the eight mediation functions (Subsection 7.5), inspects only the one model that cleared the structural contract under zero-shot, so its findings characterize that configuration and are not generalized to the full model set. The model-side mediation coding is, moreover, a single-coder pass under codebook v0.2; a blind second pass on the 39 model turns is future work, on a par with the not-yet-re-blinded v0.2 recoding of the human corpus, and the format-forced nature of FM01/FM03 in the model's output schema means the informative signal lies in the absent functions (FM02/FM04/FM06) rather than in the present ones. **Genre labeling as an uncontrolled factor.** Several scenarios open with a discursive-genre label (review, summary, retelling, opinion piece, chronicle), which for an experienced teacher can activate genre-specific evaluation criteria, such as generic adequacy, communicative purpose, and compositional organization, beyond the cohesion-coherence focus the rubric targets. Genre is not crossed with the target phenomenon in our matrix, so it is an uncontrolled factor rather than a studied variable, and some labels do not match the prototypical structure of the named genre. For the present diagnostic this is a documented limitation; a future corpus should either control genre systematically or drop the labels, and embed the target phenomena in texts closer to authentic classroom production, whose surface transparency to specialists would also be reduced. **Generalization boundary.** All results are bound to the evaluated families, their Q4 quantization, and the Ollama runtime under a zero-shot, no-coercion regime; other quantizations, runtimes, or prompting regimes may shift the structural rates, as the falsification protocol of Subsection 5.3 already shows for the one-shot case.

9 Conclusion

This study carried out a controlled diagnostic evaluation of eight open Small Language Models, run locally under infrastructure restrictions compatible with the material reality of the Brazilian public school, against a

metalinguistic rubric derived from the textual-cohesion taxonomy of Ingedore Koch. On RQ1 (native structural adherence), the structural-conformance dimension produced a sharp result: among the models in the *deployable* range (up to 3.8 billion parameters), only Qwen 2.5 3B-Instruct combines full adherence to the JSON contract declared in natural language with a mean latency comfortably below the Human–Computer Interaction threshold for synchronous tasks and low variability across repetitions. It is the only configuration evaluated that satisfies, at once and without recourse to external grammatical coercion, all the non-functional requirements for educational deployment.

On RQ2 (mobilization of the metalinguistic rubric), however, this result does not warrant an optimistic inference about the model's pedagogical adequacy. A corroborating qualitative probe shows that, although structurally conformant, the model exhibits only partial and unstable adherence to Koch's metalinguistic rubric, mobilizing it in about half of the runs (51.3%, $n=39$) and, more tellingly, doing so unstably: a single scenario out of thirteen elicits that mobilization across all three repetitions. We read this probe as corroborating the structural diagnosis, not as an independent claim of equal evidentiary weight.

Against the model's superficiality, the specialist human reference of Section 7 supplies the positive counterpart: a descriptive repertoire of eight mediation functions, reliably coded ($\kappa = 0.83$), in which what distinguishes expert feedback is not the detection of the planted problem but a small operational core (naming the problem, offering a cue, proposing the revision) surrounded by four differentiators, recognition, reflection, partial modeling, and the challenge-plus-autonomy layer, that vary across teachers into distinct mediation profiles. This repertoire reframes the pedagogical target away from a single gold answer and toward a range of acceptable mediation profiles, and it makes explicit what a specialized model would have to learn to reproduce, in complementarity to the teacher, beyond merely respecting the structural contract. Coding the conformant model against this same repertoire confirms the diagnosis empirically rather than by assertion: its mediation collapses onto format-forced recognition and open expansion (FM01, FM07) and almost never enters the expert corrective core (naming the problem 3%, offering a cue 15%, proposing a revision 0%, against 80%/69%/57% among the teachers), at times even praising as a strength the very deviation planted as the defect. It is precisely this mediational layer, not problem detection, that the off-the-shelf models leave unaddressed.

This finding, together with the architectural findings of Section 5 and the falsification protocol of Subsection 5.3, supports the central conclusion of the study: under a zero-shot regime, the open language models currently available off the shelf, within their range of computational viability for Brazilian public schools, are insufficient for writing-feedback tutoring in Brazilian Portuguese grounded in a fine metalinguistic rubric. The insufficiency takes two forms: (i) a regime of zero-shot structural non-adherence, exhibited by the Llama 3.2 and Phi-3 families, in which the model violates the syntactic-typological contract even under an explicit natural-language specification (for the Llama 3.2 family this is an obedience bias demonstrably reversible by minimal contextual anchoring (*one-shot*), and so distinguishable from compositional incapacity); and (ii) a regime of structural adherence with pedagogical superficiality, exemplified by Qwen 2.5 3B-Instruct, in which the model respects the format but operates predominantly in a thematic-surface layer, without systematically mobilizing the theoretical taxonomy the task requires.

That the structural regime persists across models from architecturally distinct families, trained by different institutions and optimized for predominantly Anglophone markets, and that the single model clearing it still fails to mobilize the rubric reliably, converges on the diagnosis that writing-feedback tutoring in Brazilian Portuguese for basic education is not solved in a zero-shot regime by off-the-shelf generalist models. It calls instead for specialized models, developed through supervised fine-tuning on curated pedagogical corpora in Brazilian Portuguese and aligned with the metalinguistic rubric specific to the school domain. The alternative route, *prompt engineering* with demonstrative anchoring, made viable by the reversibility documented in Subsection 5.3, remains technically available, but it presupposes an operating regime whose costs in context budget and latency must be weighed against the savings in training. That development lies outside the scope of the present diagnosis and constitutes the next stage of this research agenda. Concretely, the model selected here, Qwen 2.5 3B-Instruct, is the base adopted for the classroom pilot of the broader project: feedback collected during the pilot, together with curated synthetic scenarios aligned to the mediation rubric, will supply the

supervised fine-tuning corpus for a specialized writing-feedback model, which we intend to release openly so that the diagnostic benchmark and the resulting model can be cited and reused together.

A Statistical Rationale

This appendix details the inferential framework summarized in Subsection 3. The categorical comparisons that support the structural-conformance claims of the paper, namely that conformance rates differ from a reference model and that *one-shot* demonstration reverses the Llama 3.2 family bias, rely on Fisher's exact test on 2×2 contingency tables, complemented by Wilson 95% confidence intervals for each observed proportion. Fisher's exact test was chosen for four converging reasons: (i) the structural-conformance data is intrinsically binary (conformant vs. divergent); (ii) cells with extreme proportions (0/39 and 39/39) are observed, in which the asymptotic χ^2 approximation is invalid (expected frequency below 5 in at least one cell); (iii) the test produces exact two-sided *p*-values by combinatorial enumeration of more extreme tables, with no distributional assumption; and (iv) when a cell of the contingency table is zero, the conditional maximum-likelihood odds ratio is not finitely estimable, but the *p*-value derived from the exact hypergeometric distribution remains valid as the reportable effect, in line with standard practice for categorical comparisons under extreme proportion regimes.

Wilson 95% CIs supplement each rate with its uncertainty band, computed analytically without assuming normality of the proportion, which matters especially near the extremes of the unit interval, where the normal approximation behind the Wald interval would produce bounds outside [0, 1]. All inferential computations were performed with `scipy.stats.fisher_exact` (two-sided alternative) and a closed-form Wilson interval implementation, both reported in `src/inferential_statistics.py`. Significance is reported in APA-style notation throughout the paper: $p < 0.001$ ***, $p < 0.01$ **, $p < 0.05$ *, $p \geq 0.05$ n.s., with $\alpha = 0.05$ as the conventional rejection threshold.

B Implementation Details

This appendix documents the implementation artifacts referenced in Subsections 3.3 and 3.4. The materials are provided to support integral reproducibility of the study; the executable form, together with the raw inference corpus, is available in the public repository indicated in Section 3.

B.1: HTTP payload sent to Ollama. All 312 main-protocol calls and the 80 falsification-protocol calls sent the server a JSON payload built in a rigorously uniform way across models, reproduced schematically below:

```
payload = {
  "model": <model identifier>,
  "messages": [
    {"role": "system", "content": SYSTEM_PROMPT},
    {"role": "user", "content": <scenario text>},
  ],
  "stream": False,
  "format": "json",
  "options": {"temperature": 0.2},
}
```

Each hyperparameter was selected by explicit methodological motivation and acts on a distinct dimension of the generation process's stochasticity:

1. **temperature = 0.2.** The parameter scales logits prior to *softmax* sampling and governs the concentration of the probability distribution over the vocabulary. Values close to 0 collapse the distribution into a near-deterministic regime (quasi-greedy prediction); values above 1 flatten it, favoring lexical diversity. The choice of 0.2 establishes a compromise: rigorous enough to induce adherence to the structural contract declared in natural language by the *system prompt*, without suppressing the micro-variability necessary for meaningful estimation of intra-model CV across repetitions. Values near 0 would make it impossible to separate systemic from stochastic variance, the central axis of the instruction-stability analysis.

2. `stream = false`. Deactivates the progressive *token-by-token* transmission regime (*server-sent events*) and imposes synchronous mode, in which the server retains the entire response until stopping and returns it in a single HTTP body. The justification is metrological: the *wall-clock* latency meter based on `time.time()` requires well-defined start and end points, and reception in a single chunk eliminates ambiguity about the inference termination moment. The synchronous mode is also the regime expected in real school usage, in which the student awaits the complete response before proceeding.
3. `format = "json"`. Signals to Ollama the intent of JSON output. Operationally, the server restricts the decoder to a logit mask compatible with tokens structurally valid for JSON, exerting a probabilistic *nudging* over the generation. This is a critical methodological point: the parameter does not enforce a schema. It only guarantees that the output is syntactically parseable as JSON, and imposes nothing on field typology, the presence of required keys, or list cardinality. Keeping it active, without complementing it with strict grammatical coercion (BNF grammars, regex validation, or *constrained decoding*), reflects the intent to gauge the model's native adherence to the structural contract specified in natural language, under conditions representative of a realistic school-deployment scenario, where there are no engineering resources to add layers of syntactic coercion to every inference.
4. `timeout = 120` seconds. Client-side waiting limit, dimensioned to accommodate even the most prolonged generations of verbose models without masking genuine server failures.

B.2: Source code of the `divergente()` structural validator. Each response was classified for conformance by the deterministic function below, implemented in `src/mine_benchmark.py`, which decides, from the raw content returned in `message.content`, whether the response fully satisfies the contract declared in the *system prompt*. The classifier logic is reproduced below, in the exact order of execution applied over the 312-record corpus:

```
def divergente(raw_response: str) -> bool:
    # C1 - JSON syntax
    try:
        parsed = json.loads(raw_response)
    except json.JSONDecodeError:
        return True

    # C2 - Root type
    if not isinstance(parsed, dict):
        return True

    # C3 - Type and cardinality of `pontos_fortes`
    pf = parsed.get("pontos_fortes")
    if not (isinstance(pf, str) and pf.strip()):
        return True

    # C4 - Type and cardinality of `perguntas_reflexivas`
    pr = parsed.get("perguntas_reflexivas")
    if not (isinstance(pr, list) and len(pr) > 0 and
            all(isinstance(x, str) and x.strip() for x in pr)):
        return True

    return False
```

The function operates on the raw content preserved in the `resposta_ia` attribute of each of the 312 records of the collection corpus. The four conditions (C1–C4) are documented in narrative form in Subsection 3.4.

Declarations

Author Contributions

Following the CRediT Taxonomy (<https://credit.niso.org/>): **Randerson O. M. Rebouças**: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing (original draft). **Marcelo Magalhães Foohs**: Supervision, Validation, Writing (review & editing). **Rosa Maria Vicari**: Supervision, Conceptualization (pedagogical-linguistic rubric), Validation, Writing (review & editing). All authors read and approved the final manuscript.

Competing Interests

The authors declare that they have no competing interests.

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Availability of Data and Materials

All materials required to reproduce this study are publicly available. The source code (collection, classification, aggregation, and inferential-statistics scripts) is released under the MIT License, and the data (the 13 canonical scenarios, the 312-record primary inference corpus, and the 80-record falsification corpus) under the Creative Commons Attribution 4.0 (CC BY 4.0) License. A permanent, citable archive of the release is deposited on Zenodo under the concept DOI 10.5281/zenodo.20388846, which always resolves to the latest version, with the development repository hosted at <https://github.com/RandMelville/slm-writing-feedback-tutor-ptbr>.

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